

Your grade will be notified, and you will be able to access the feedback to your answers,

at 7:00pm of Wednesday, March 31, 2021.

1.

Consider a country consisting of the goods market only, where $C = c_0$, $G = \bar{G} - gY$, $T = \bar{T}$, and $I = \bar{I}$, with g a positive constant and the remaining symbols having the usual meaning. Write the expression of the multiplier for this economy, and then assume that government purchases of goods and services and net taxes are raised by the same amount and at the same time, so that $\Delta\bar{G} = \Delta\bar{T} > 0$. By making explicit reference to the goods markets equilibrium condition written as an equality between national saving and investment, explain if, and why, it is possible to say something about the direction in which private saving will have changed in the new equilibrium [N.B.: you are not expected to derive the analytical expression for the change in private saving, but just to explain whether it is possible to conclude that, in the new equilibrium, it will have gone up, gone down, or remained constant].

SKETCH OF ANSWER

The multiplier – the constant that, in the expression for equilibrium income, multiplies autonomous spending (in this economy given by the sum of autonomous consumption, investment, and the autonomous component of government purchases of goods and services) – is equal to 1 over $(1+g)$, a positive quantity. If $\Delta\bar{G} = \Delta\bar{T} > 0$, income will be higher in the new equilibrium (the increase in net taxes has no impact on the demand for goods, since this latter does not depend on $\bar{T}$, while that in $\bar{G}$ of course raises it). In the new equilibrium, national saving will have to remain unchanged, just like investment (which is entirely exogenous). Since public saving increases (being $\Delta\bar{G} = \Delta\bar{T}$, it will change only because of the fall in G due to the increase in income), for national saving to remain constant private saving must have gone down by the same amount by which public saving has gone up.

2.

Consider an economy with constant prices described by an “extended” IS-LM model, with a central bank that chooses the interest rate. How will a decrease in the degree of risk aversion prevailing in the country’s financial markets affect the position in the plane and/or the slope of the curves represented in the IS-LM diagram? Say the central bank is determined to keep income constant even in face of this change. To achieve its aim, should it buy or sell bonds in the open market? And what if it decides to prevent income from varying by means of a change in the reserve ratio θ ? In this case, should it try to cause an increase or a decrease in θ ? Why? Explain.

SKETCH OF ANSWER

In the “extended” IS-LM model, investment depends on the borrowing rate, sum of the real interest rate, r , and the risk premium, x . A decrease in the degree of risk aversion lowers the risk premium and the borrowing rate for any given level of the real interest rate. For any given r , it therefore leads to higher investment and aggregate demand for goods. The IS curve shifts to the right, and equilibrium income rises. To keep it at the initial level, the central bank must implement a contractionary monetary policy, one that – by lowering the supply of money – leads to an increase in the interest rate. To this end, it could sell bonds in the open market, or engineer an increase in the reserve ratio (something that, causing a contraction in commercial banks’ lending, and therefore in bank deposits, represents an alternative way of lowering M and raising i , thus achieving the aim of preventing income from changing).

3.

True or False?

Explain whether the following statement is true or false. Motivate your answer in a brief but rigorous way, by making explicit reference to the relevant theory. Lack of proper explanations will result in zero points.

“The zero lower bound for the nominal interest rate implies that, if expected inflation is -2% , then the real interest rate cannot be smaller than 2% ”.

SKETCH OF ANSWER

True. Being $r = i - \pi^e$, and given expected inflation, r is lowest when $i = 0$, so that $r \geq -\pi^e$. It follows that, if $\pi^e = -2\%$, the real interest rate cannot be less than 2% .

4.

Macro is an economy with flexible prices, described by a standard IS-LM-PC model and initially in a medium-run equilibrium position. The country is hit by an adverse supply shock consisting in a permanent increase in the oil prices. At the same time, Macro also experiences a permanent decrease in net taxes. Having discussed which of the curves represented in the IS-LM-PC diagram will be affected by these two changes, and how, define the concept of natural (or “neutral”, or “Wicksellian”) real rate of interest, r_n . How will have r_n changed in the new medium-run equilibrium now prevailing in Macro? Provide the economic intuition behind this change in the natural interest rate, and discuss how the composition of aggregate demand will have changed in the move from the initial to the final medium-run equilibrium.

SKETCH OF ANSWER

The rise in oil prices lowers the natural level of production and, in the IS-LM-PC diagram, shifts to the left the PC curve, that will now cross the x-axis for the new, lower level of Y_n . The cut in net taxes will instead shift the IS curve to the right. The natural real interest rate – the value of r for which the demand for goods is equal to Y_n – is now higher. In fact, and for given r_n , in the medium-run the two shocks under consideration lead to an excess demand for goods (the medium-run supply of goods has gone down, the demand up). It follows that the natural interest rate will have to go up by the amount needed to eliminate this excess demand and make sure a new medium-run equilibrium will be reached. Finally, in the move from the initial to the final medium-run equilibrium, G is of course unchanged, I has gone down (r is higher, Y is lower), while the direction of change of consumption C is uncertain (Y is lower, but net taxes are lower, too).